

CASE STUDY

HOTEL BOOKING SYSTEM

Submitted By

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1. Introduction

Overview

The **Hotel Booking System (HBS)** is a web-based software application developed to automate and simplify the hotel reservation process. It enables customers to search for available rooms, make reservations, modify or cancel bookings, and receive booking confirmations through an online platform. At the same time, it provides hotel administrators with a centralized system to manage room inventory, customer information, bookings, and reports efficiently.

Traditionally, hotel reservations have relied on manual record-keeping, phone calls, or in-person bookings, which are often time-consuming and susceptible to human errors such as double bookings and inaccurate records. The proposed Hotel Booking System addresses these limitations by providing a secure, reliable, and user-friendly solution that improves operational efficiency and customer satisfaction.

Objectives

The objectives of the proposed system are to:

- Automate the hotel reservation process.
- Provide real-time room availability.
- Simplify room booking and cancellation.
- Improve data accuracy and operational efficiency.
- Generate reports to support hotel management.

2. Functional Requirements

Customer

- Register/Login
- Search Rooms
- View Room Details
- Book Room
- Cancel Booking
- View Booking History

Administrator

- Manage Rooms
- Manage Bookings
- Generate Reports
- Manage Customer Records

3. User Stories

Customer

- **As a customer,** I want to search for available rooms so that I can choose a room suitable for my travel dates.
- **As a customer,** I want to book a room online so that I can reserve my accommodation before arrival.
- **As a customer,** I want to cancel my booking so that I can update my travel plans if necessary.
- **As a customer,** I want to view my booking history so that I can keep track of my previous and upcoming reservations.

Administrator

- **As an administrator,** I want to manage room details so that room information remains accurate and up to date.
- **As an administrator,** I want to manage customer bookings so that reservations are processed efficiently.
- **As an administrator,** I want to generate booking reports so that I can monitor hotel occupancy and business performance.
- **As an administrator,** I want to manage customer accounts so that customer information remains accurate and up to date.

4. Software Development Process Model

Agile Development Model

The **Agile Development Model** is selected for this project because it supports iterative development, continuous testing, and frequent customer feedback. Instead of developing the entire system at once, Agile divides the project into smaller development cycles known as **sprints**. Each sprint delivers a functional component of the system, allowing improvements to be incorporated throughout the development process.

Sprint	Activity
Sprint 1	Gather and analyze user requirements.
Sprint 2	Design the system architecture, database, and user interface.
Sprint 3	Develop user authentication, room management, and booking modules.
Sprint 4	Integrate payment functionality and perform system testing.
Sprint 5	Deploy the application and collect user feedback.
Sprint 6	Maintain the system by fixing bugs and implementing enhancements.

Advantages of Agile

- Faster delivery of functional software.
- Continuous customer feedback.
- Easy adaptation to changing requirements.
- Improved software quality through frequent testing.

5. Programming Language & Technology Stack

Technology	Purpose
Programming Language	Java – Used to implement the core application logic using object-oriented programming principles.
Frontend	HTML, CSS, JavaScript – Used to develop a responsive and interactive user interface.

Backend	Spring Boot – Handles business logic, REST APIs, and server-side processing.
Database	MySQL – Stores customer, room, booking, and payment information securely.
Version Control	GitHub – Manages source code versions and supports collaborative development.

6. Implementation Strategy

The implementation of the Hotel Booking System follows a structured and incremental approach.

Requirement Analysis

Gather functional and non-functional requirements from customers and hotel administrators to define the project scope.

System Design

Design the database schema, user interface, and application architecture using UML diagrams.

Development

Develop the application modules, including user authentication, room management, booking management, and payment processing.

Testing

Test each module individually before integrating the complete system to ensure functionality and reliability.

Deployment

Deploy the application to the production environment and make it accessible to users.

Maintenance

Monitor system performance, resolve software defects, and introduce new features based on user feedback.

7. Evaluation Strategy

The Hotel Booking System is evaluated using the following software testing techniques to ensure quality and reliability.

Testing Method	Description
Functional Testing	Verifies that all system features operate correctly according to requirements.
Performance Testing	Evaluates system speed, responsiveness, and stability under different workloads.
Security Testing	Ensures data security and prevents unauthorized access.
Usability Testing	Ensures the system is easy to use and provides a positive user experience.
User Acceptance Testing (UAT)	Verifies that the system meets user requirements before deployment.

8. UML Diagrams

8.1 Use Case Diagram of the Hotel Booking System

Actors

- Customer
- Administrator

Customer Use Cases

- Register/Login
- Search Rooms
- View Room Details
- Book Room
- Cancel Booking

Administrator Use Cases

- Manage Rooms
- Manage Bookings
- Generate Reports

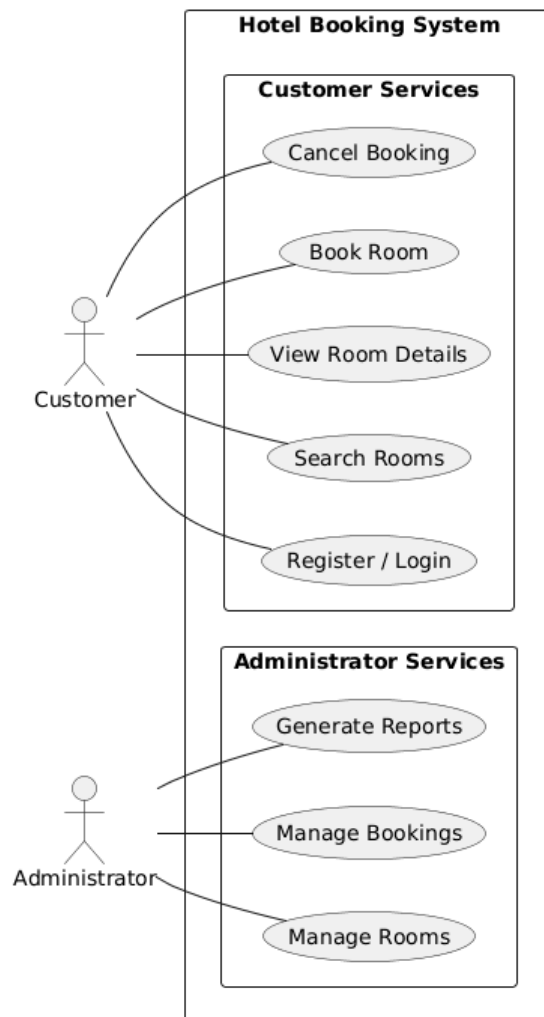


Figure 8.1 illustrates the interaction between the Customer and Administrator within the Hotel Booking System. Customers perform booking-related operations, while administrators manage rooms, bookings, and reports.

8.2 Class Diagram of the Hotel Booking System

Classes

Customer

- CustomerID
- Name
- Email
- Phone
- Login()
- BookRoom()

Booking

- BookingID
- CheckInDate
- CheckOutDate
- BookingStatus
- TotalAmount

Room

- RoomID
- RoomType
- Price
- Availability

Administrator

- AdminID
- Name
- ManageRooms()
- ManageBookings()
- GenerateReports()

Relationships

- A **Customer** creates one or more **Bookings**.
- Each **Booking** is associated with one **Room**.
- The **Administrator** manages **Rooms** and **Bookings**.

Hotel Booking System - Class Diagram

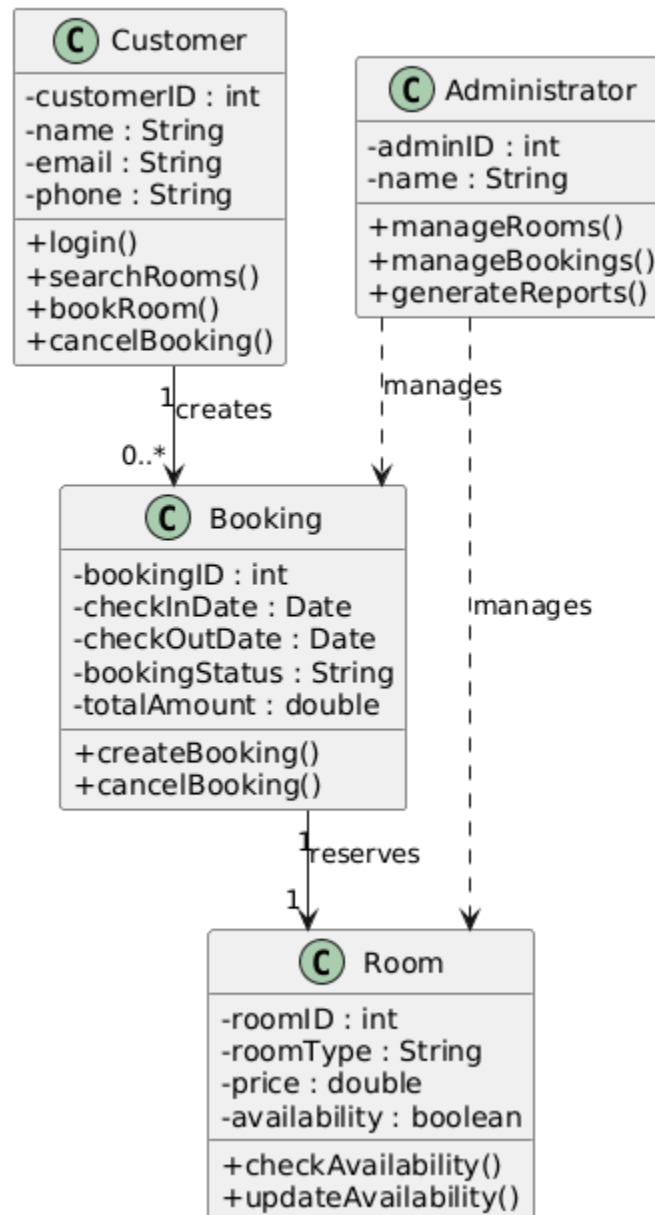


Figure 8.2 presents the static structure of the Hotel Booking System. It illustrates the main classes, their attributes, methods, and relationships. Customers create bookings, bookings are associated with rooms, and administrators manage both rooms and bookings.

9. Expected Outcomes

The proposed Hotel Booking System is expected to achieve the following outcomes:

- Faster and more efficient room booking.
- Improved customer experience through online reservations.
- Reduced manual effort for hotel administrators.
- Accurate room availability management.
- Secure and reliable booking records.
- Better operational efficiency through automated reporting.

10. Conclusion

The proposed **Hotel Booking System** provides an efficient and reliable solution for managing hotel reservations in a digital environment. By automating room booking, reservation management, and administrative operations, the system minimizes manual effort, improves operational efficiency, and enhances customer satisfaction.

The adoption of the **Agile Development Model** ensures flexibility throughout the software development lifecycle, while the selected technology stack—**Java, Spring Boot, MySQL, HTML, CSS, and JavaScript**—provides a scalable, secure, and maintainable platform. The system demonstrates the practical application of software engineering principles to solve real-world business challenges and can be further enhanced with features such as online payment gateways, mobile applications, AI-based room recommendations, and customer review management.

References

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